# [4+4] cycloaddition | orbital phases along the reaction

## REACTANT ORBITALS

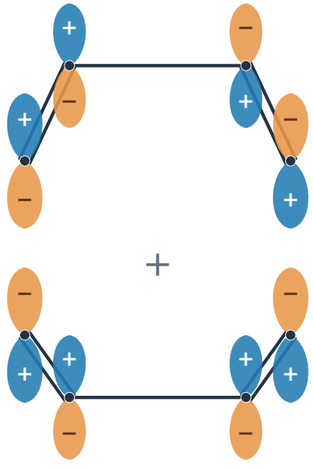
## PROPOSED TRANSITION STATE

## PRODUCT

### THERMALLY ALLOWED (FORMAL)

suprafacial / antarafacial

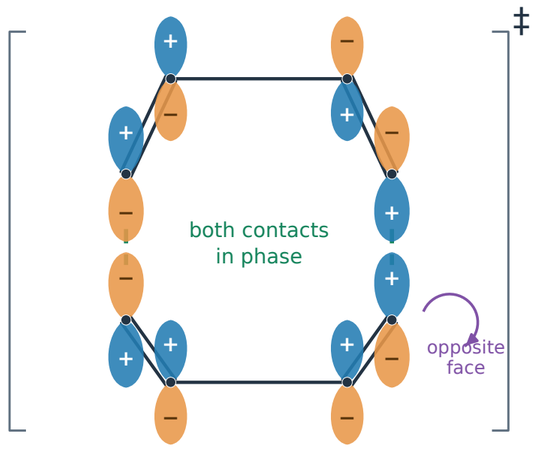
Diene HOMO, ψ₂ (occupied)



Diene LUMO, ψ₃ (empty)

Δ

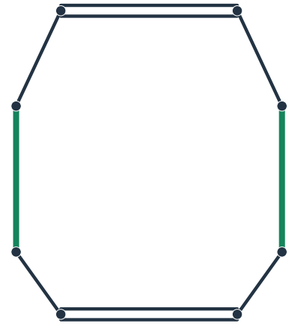
Möbius circuit · 8 electrons




Partial bonds join the facing lobes.

Δ

Product connectivity



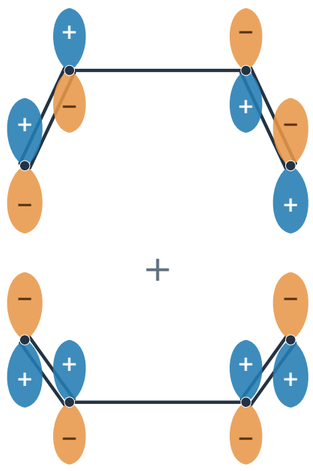
Cycloocta-1,5-diene

New σ bond(s) highlighted in green.

### THERMALLY FORBIDDEN

suprafacial / suprafacial

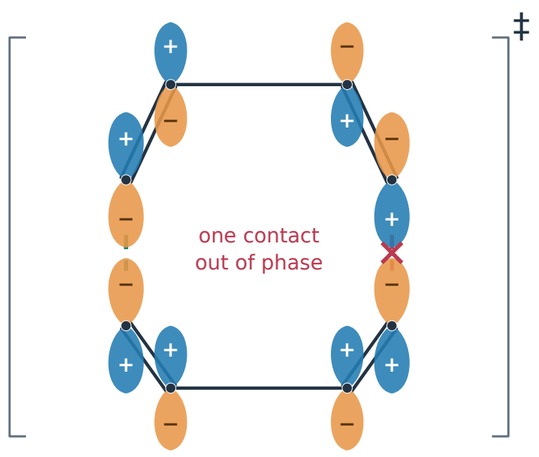
Diene HOMO, ψ₂ (occupied)



Diene LUMO, ψ₃ (empty)

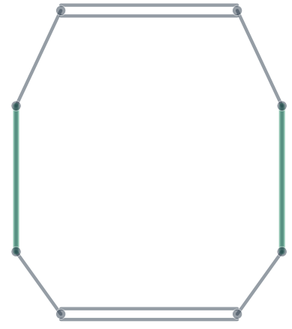
Δ

Hückel circuit · 8 electrons




Partial bonds join the facing lobes.

Proposed product connectivity



Cycloocta-1,5-diene

Forbidden for this concerted thermal route.

+ / − orbital phase (not charge)

forming overlap / bond

phase mismatch

Antarafacial allowance is formal: geometry may prevent this approach, especially for ethene. Products show connectivity only.

Thermal, concerted pathways. Colored lobes are qualitative phase schematics; the middle panels are not optimized TS geometries or TS eigenorbitals.